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Software Quality Assurance Lecture 1

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- Defining Quality
- Quality Characteristics
- Difference between Quality Assurance & Quality Control
- Six Testing Principles
- Types of Testing
- Testing throughout the Software Lifecycle

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Defining 'Quality'

- *"Meeting the requirements of the customer."*
- *"The standard of something as measured against other things of a similar kind; the degree of excellence of something"*
- *"Quality is the degree to which a commodity meets the requirements of the customer at the start of its life. (ISO 9000)"*

Who is Responsible for 'Quality' ?

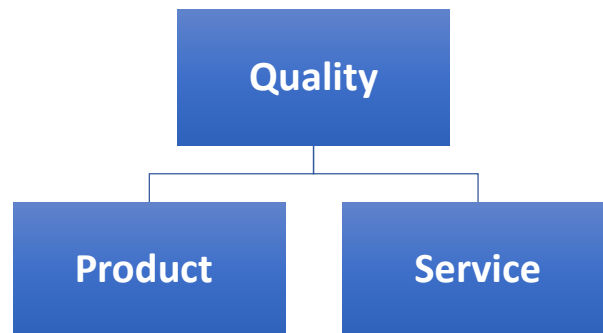


The 'Team' is Responsible for 'Quality'



Quality Characteristics

Quality Characteristics



Quality Characteristics : Product Quality Dimensions

- The ISO 9126-1 software quality model identifies **6 main quality characteristics**, namely:
 - Functionality
 - Reliability
 - Usability
 - Efficiency
 - Maintainability
 - Portability

Quality Characteristics

Product Quality Dimension - **Functionality**

Sub – Characteristic	Definition
Suitability	This is the essential Functionality characteristic and refers to the appropriateness (to specification) of the functions of the software
Accurateness	This refers to the correctness of the functions
Interoperability	A given software component or system does not typically function in isolation. This sub characteristic concerns the ability of a software component to interact with other components or systems
Compliance	Where appropriate certain industry (or government) laws and guidelines need to be complied
Security	This sub characteristic relates to unauthorized access to the software functions.

Quality Characteristics

Product Quality Dimension - **Reliability**

Sub – Characteristic	Definition
Maturity	This sub characteristic concerns the frequency of failure of the software. It means the attributes of software that bear on the frequency of failure by faults in the software
Fault tolerance	The ability of software to withstand (and recover) from component, or environmental, failure
Recoverability	Ability to bring back a failed system to full operation, including data and network connections

Quality Characteristics

Product Quality Dimension - Usability

Sub – Characteristic	Definition
Understandability	Determines the ease of which the systems functions can be understood, relates to user mental models in Human Computer Interaction methods
Learnability	The capability of the software product to enable the user to learn its applications. Learning effort for different users, i.e. novice, expert, casual etc.
Operability	Ability of the software to be easily operated by a given user in a given environment
Attractiveness	The capability of the software product to be liked by the user.

Quality Characteristics

Product Quality Dimension - Efficiency

This characteristic is concerned with the system resources used when providing the required functionality

Sub – Characteristic	Definition
Time behavior	The capability of the software to provide appropriate response and processing times and throughput rates when performing its function under stated conditions.
Resource Utilization	The capability of the software to use appropriate resources in an appropriate time when the software performs its function under stated conditions. i.e., memory, CPU, disk, and network usage.

Quality Characteristics

Product Quality Dimension - Maintainability

The ability to identify and fix a fault within a software component

Sub – Characteristic	Definition
Analyzability	The capability of the software product to be diagnosed for deficiencies or causes of failures in the software or for the parts to be modified to be identified.
Changeability	The capability of the software product to enable a specified modification to be implemented.
Stability	The capability of the software to minimize unexpected effects from modifications of the software.
Testability	The capability of the software product to enable modified software to be validated.

Quality Characteristics

Product Quality Dimension - Portability

Software can adopt to changes in its environment or with its requirements

Sub – Characteristic	Definition
Adaptability	The capability of the software to be modified for different specified environments without applying actions or means other than those provided for this purpose for the software considered.
Installability	The capability of the software to be installed in a specified environment.
Replaceability	Characterizes the <i>plug and play</i> aspect of software components, that is how easy is it to exchange a given software component within a specified environment.
Coexistence	The capability of the software to coexist with other independent software in a common environment sharing common resources.

Quality Characteristics

Service Quality Dimension

Characteristic	Definition
Tangibles	Appearance of physical facilities, equipment, personnel, and communication materials
Reliability	Ability to perform the promised service dependably and accurately
Responsiveness	Willingness to help customers and provide prompt service
Assurance	Knowledge and courtesy of employees and their ability to convey trust and confidence
Empathy	The caring, individualized attention the firm provides its customers

Testing Example -01



Testing Example -02



Quality Control (QC)

- Is a set of measures taken to ensure that defective products or services are not produced, and that the design meets performance requirements
- The function of software quality that checks that the project follows its standards, processes, and procedures, and that the project produces the required internal and external (deliverable) products
- Building the right product

Quality Assurance (QA)

- The function of software quality that assures that the standards, processes, and procedures are appropriate for the project and are correctly implemented-----
- Building the product right

QA Vs. QC

QA is the process of managing for quality

Prevent defects from entering into the solution in the first place



QC is used to verify the quality of the output

reactive means by which quality is gauged and monitored

What is a 'Defect' / 'Bug' ?



What is a 'Defect' / 'Bug' ?

- A human being can make an error (mistake), which produces a defect (fault, bug) in the ***program code or in a document***. If a defect in code is executed, the system may fail to do what it should do (or do something it shouldn't), causing a failure. Defects in software, systems or documents may result in failures, **but not all defects do so**.
- Defects occur because human beings are fallible and because there is **time pressure, complex code, complexity of infrastructure, changing technologies, and/or many system interactions**

Popular Software Test Failures

- **Ariane 5 Explosion[9/1997] - \$500 million**

Computer program trying to stuff a 64-bit number into a 16-bit space

- **NASA Mars Climate Orbiter [9/23/1999] - \$125 million**

It was a mathematical mismatch (metric / non- metric)

- **Top 10 Bugs** <https://www.youtube.com/watch?v=AGI371ht1N8>

Airline Crash



264 People dead



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Why is Testing Important?

Software Bugs can potentially
cause Monetary or even loss of life



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Therac-25 Radiation Therapy Machine failure



- 1985
- For six patients in 1986 and 1987, it exposed them to massive overdoses of radiation, killing four and leaving two others with lifelong injuries.



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Thank You



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